

Quantum Mechanics III

Homework 11

Seonwoo Park

Due 2026-11-15 23:59

Topic

Parameter Estimation and Experimental Design

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Parameter Estimation and Experimental Design. State all assumptions and conventions.

$$F_{ij} = E[\partial_i \ln p \partial_j \ln p]$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$F_Q = \text{Tr}(\rho L_i L_j)$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$x_{\text{next}} = \text{argmax}_x E_D[KL(p(\theta|D, x)||p(\theta|D))]$$

Show enough intermediate work that another student can reproduce the calculation.